

SUBHAJIT DAS

AI Engineer | Generative AI Engineer | RAG, Hybrid Retrieval, LLM Evaluation

(+91) 9903010699

dassubhojit750075@gmail.com

<https://github.com/Subhajitdas99>

[linkedin.com/in/subhajit-das-8207ab240/](https://www.linkedin.com/in/subhajit-das-8207ab240/)

Summary

AI Engineer with hands-on experience building retrieval-augmented generation (RAG) systems, hybrid search pipelines, and production-style ML applications. Skilled in Python, FastAPI, FAISS, BM25, LangChain, Hugging Face Transformers, Sentence-Transformers, TensorFlow, Docker, and MLflow. Built document intelligence and machine learning systems with a focus on retrieval quality, hallucination reduction, latency optimization, and scalable API deployment. Strong interest in designing real-world AI systems that combine LLMs, search, evaluation, and reliable production workflows.

EDUCATION

B.Tech in Computer Science & Engineering (AI & ML), (Pursuing)|2023-2026

College:Dr. Sudhir Chandra Sur Institute of Technology and Sports Complex, Dumdum, Kolkata

CGPA: --

Diploma in Electrical Engineering (WBSCTE) | Percentage: 72.4% | Year: 2021

Higher Secondary (XII) | Percentage: 63.2% | Year: 2018

Secondary(X) | Percentage: 70.1% | Year: 2015

KEY SKILLS

Languages & Frameworks: Python, FastAPI, TensorFlow, Keras, Scikit-learn, Pandas, NumPy

Generative AI & NLP: RAG Pipelines, LangChain, HuggingFace Transformers, Sentence-Transformers, Vector Embeddings, Prompt Engineering

Computer Vision: OpenCV, CNNs, YOLO, Real-time Video Inference

Machine Learning: XGBoost, Feature Engineering, Model Evaluation, Imbalanced Learning, Data set handling, Model training, Evaluation metrics,

Retrieval & Search: FAISS, BM25 (rank-bm25), Hybrid Retrieval, Semantic Search, Vector Embeddings

Evaluation & Monitoring: Recall@K, Retrieval Evaluation, Faithfulness, Hallucination Reduction, Latency Tracking, MLflow

MLOps & Deployment: Docker, MLflow, REST APIs, Git/GitHub, Streamlit, Uvicorn

Concepts:LLMEvaluation,HallucinationReduction,Tool-AugmentedPipelines,Production AI Architecture

Tools: Git/GitHub, Jupyter Notebook, Google Colab, VS Code, Docker, MLflow, Streamlit

Soft Skills: Leadership, Problem Solving, Time Management, Team Building, Critical Thinking, Product Management

Projects

Real-Time CCTV Analytics — Object Detection & Tracking Pipeline

Tech stack – YOLO, OpenCV, Python, FastAPI, Docker

- Built a real-time object detection pipeline using YOLO for live video streams
- Designed modular inference workflow for multi-object detection
- Implemented real-time video frame processing and detection overlay
- Structured pipeline for production-style deployment
- Integrated detection pipeline with API serving layer for scalable inference
- Focused on latency optimization and detection reliability

Face Mask Detection

- **Tech Stack:** TensorFlow, Keras, OpenCV Developed a CNN-based real-time face mask detection system
- Built dataset preprocessing and augmentation workflow
- Implemented live video inference pipeline using OpenCV
- Achieved ~90% real-time detection accuracy
- Designed end-to-end perception workflow from training to deployment
- Optimized inference performance for real-time camera feeds

Fraud Detection System

Tech: XGBoost, FastAPI, Docker, Git LFS

- Built ML classification model for fraud detection on imbalanced dataset (~140MB)
- Achieved PR-AUC 0.87 and Recall 87% via threshold tuning & imbalance handling
- Developed real-time prediction REST API using FastAPI
- Containerized ML system using Docker
- Built reproducible ML pipeline with versioned datasets

Enterprise GenAI RAG Assistant

Tech Stack: Python, FastAPI, Uvicorn, Streamlit, Sentence-Transformers, FAISS, BM25 (rank-bm25), HuggingFace Transformers (FLAN-T5), MLflow, Docker, Git/GitHub.

- Built an Enterprise Document Q&A (RAG) system to upload PDFs and query them via a conversational interface.
- Implemented full retrieval pipeline using FAISS (semantic search), BM25 (keyword search), and a Hybrid Retriever with score normalization.
- Developed a FastAPI backend with endpoints for document ingestion, indexing, and query answering, plus response sources/page tracing.
- Added MLflow-based evaluation & monitoring to track metrics like Recall@K, Faithfulness, Hallucination Rate, and Latency for retriever A/B testing.
- Created a Streamlit UI for file upload, chat-style interaction, and transparent source inspection for answers.

EXPERIENCES

Supervisor – Joy Guru Eng. Works (Renuka Sugar Pvt. Ltd.) | Apr 2023 – Aug 2023 • Led and supervised repair operations including pipe blasting, instrumentation maintenance.

• Managed a small team and executed field calculations, team coordination, and reporting.

Associate – Manpower Group at Adani Solar | Dec 2021 – Mar 2023 • Worked in solar panel manufacturing and production environments. • Focused on machine handling, safety compliance, and quality control.

CERTIFICATIONS

Google Crash Course on Python (<https://coursera.org/verify/4PBTEB35MS2W>)

Internship (AI & Cyber Security) – SurTech Research & Innovation Center (3 weeks)

The Introduction to Data Science Course – Cisco Networking Academy

Certificate of Artificial Intelligence Fundamental – IBM (Nov 2025)

Artificial Intelligence & Machine Learning Training — Internshala Trainings - Duration: 6 weeks | Certified on: Oct 2025

n8n – No Code AI Agent Builder | Certification (Dec 2025)

IBM Granite Models for Software Development - Certification (March - 2026)